

Supplement to: Conditional prediction of source-maintained lattice fields

Emergent Dynamics Laboratory

V2 — 4 September 2026

This supplement accompanies the revised source-response manuscript. The main claim is conditional point compatibility. Reconstruction uses existing bytes and introduces no engine or predictor simulations.

S1 Exact protocol and implementation

The authoritative protocol is `OBTC02/code/obtc02_protocol.yaml`; the fresh allocation and decision rule are in `OBFOR01/out/_freeze.json`. At each directed transport pass, the offered count is binomial and the accepted count is clipped to destination free capacity, using the state left by the preceding pass. Reaction uses local XY , substrate availability and capacity. X and Y decay into their respective waste species. The balanced exchange layer is implemented in `ORR01/code/lawspec_v2.py`, not by a continuum approximation.

The exact source chain supplied with the package is:

- `ORR01/code/kinetics.py`, `lawspec_v2.py`, `observe.py`: physical update, reservoir exchange and recorder;
- `OBTC02/code/engine_obtc.py`, `protocol_obtc02.py`, `metrics_obtc.py`, `guard_obtc.py`: seeded world, protocol and observable;
- `OBFOR01/code/run_obfor01.py`: fresh runner and sub-step ledgers;
- `OBFOR01/code/m6_obfor01.py`: conditional predictor and ablation switches;
- `SEAL01/code/seal_repair.py` and `seal_flow_and_numbers.py`: historical seal calculations, preserved as source evidence.

Names without directory prefixes inherit the preceding directory. Legacy scripts retain original container paths and are not the V2 portable entry points. The V2 pipeline is `code/reproduce.py`; it operates on the included raw records and does not invoke legacy engine or predictor runners.

The fresh seeds are 9300000–9300013 for the static condition and 9300014–9300027 for the mobile condition. The original freeze excludes all recorded retired seeds. There are 220 sampled frames in an arm; the 180 with step greater than 2000 enter the analysis. All frames and ledgers are checked for shape, and frame steps for cadence and completeness. The median uses the discrete radius about the source, not about a fitted centre.

The source starts with the frozen seeding routine. The physics generator, feed generator and diagnostic tracker have the distinct roles recorded in the source code. No diagnostic particle identifier affects the physical rates. Original paths remain in the immutable source tree rather than being silently rewritten.

S2 Provenance and chronology

The restored scientific tip is 06c592313df96601de8d2a89676d5a5cf79fc414. The original LRCPS01 paper package is preserved at 0a872ac; V2 changes its interpretation, not the engine or raw data. The pre-run OBFOR01 freeze is 050e666ed40dbcc18cc26d87e2e4b17b7c096ed6, and the first fresh-arm commit is 0148acc7dc52d10d0a0d62cd6048f7a8624462. The freeze is an ancestor of the raw-data commit, and fourteen listed load-bearing files have the same Git blobs at the freeze and restored tip. See `provenance/GIT_FREEZE_VERIFICATION.json`.

This demonstrates recorded pre-run ordering, not an external registration timestamp. The governing historical run-budget mandate is absent, so its compliance is **UNKNOWN**, not certified and not asserted violated. The observer-inertness record compares one seed in plain and instrumented engines for 1 500 steps and reports equal final states. V2 verifies that record’s source binding but does not rerun the engine fixture. It is not a proof of all possible observer trajectories.

S3 All fresh arms

This table is generated from the raw NPZ files. Each row contributes one independent unit to its condition mean. Full per-arm digests and checks are in `provenance/FRESH_ARM_RECOMPUTATION.csv`.

Condition	Seed	Median r_{80}	Mean r_{80}	Window N_X	Blocked X
M	9300014	8.0000	8.3372	121.53	0.0004749
M	9300015	8.3657	9.4336	109.86	0.0003477
M	9300016	7.8102	8.0010	119.10	0.0002948
M	9300017	8.2462	8.4184	123.02	0.0004354
M	9300018	8.0623	8.0076	122.37	0.0003855
M	9300019	7.6158	8.1442	117.82	0.0003948
M	9300020	8.0623	8.3475	122.83	0.0002681
M	9300021	8.0623	8.4545	125.84	0.0004640
M	9300022	8.0623	8.2617	124.65	0.0003512
M	9300023	8.5440	8.8714	118.95	0.0003834
M	9300024	8.0623	8.4141	123.37	0.0004215
M	9300025	8.0623	8.1322	120.70	0.0004360
M	9300026	8.0623	8.3641	118.82	0.0003734
M	9300027	8.0623	8.4575	127.22	0.0005086
S	9300000	6.0828	6.0829	83.04	0.0003672
S	9300001	5.8310	5.8507	82.80	0.0003327
S	9300002	6.0828	6.0403	81.73	0.0005970
S	9300003	6.0828	6.2374	81.13	0.0002996
S	9300004	6.0000	5.9451	81.72	0.0002612
S	9300005	5.9155	5.9153	78.35	0.0004285
S	9300006	6.0828	6.0964	83.68	0.0004793
S	9300007	6.0000	5.9363	80.33	0.0004818
S	9300008	6.0828	6.1510	83.77	0.0003282
S	9300009	6.0000	5.9409	82.99	0.0003544
S	9300010	5.8310	5.9303	82.48	0.0004335
S	9300011	6.0828	6.1416	83.21	0.0005108
S	9300012	6.0828	6.0673	82.96	0.0005431
S	9300013	5.8310	5.7675	75.87	0.0003621

S4 Decision rule, uncertainty and diagnostics

For an absolute endpoint, let $\bar{r}$ be the mean of the 14 arm medians and p its frozen prediction. The criterion of record is

$$|e| \leq 2.9, \quad e = 100(\bar{r}/p - 1).$$

Its sampling standard error is $100s/(\sqrt{14}p)$, with s the sample standard deviation across arms: 0.463% static and 0.725% mobile. The predictor dispersions from stored SEAL01 replica means are 0.315% and 0.563% on the seal’s reported scale. They describe repeated 30-arm predictor calculations, not extra engine arms.

For $R = \bar{r}_M/\bar{r}_S$, disjoint seed sets give the first-order relative standard error

$$s_{\text{rel}} = \sqrt{(s_M/(\sqrt{14}\bar{r}_M))^2 + (s_S/(\sqrt{14}\bar{r}_S))^2}.$$

On the scale $100(R/p_R - 1)$ it becomes $100(R/p_R)s_{\text{rel}}$, or 0.863%. The historical adjudication omitted R/p_R in this diagnostic scale conversion. V2 records this small correction; the frozen point result is unchanged. The reported ratio predictor dispersion 0.645% uses the inherited quadrature propagation, not an independently simulated ratio distribution.

A whole-interval rule was not frozen. The post-freeze absolute equivalence diagnostics use the smaller distance to the boundary and Student’s t upper tail with 13 degrees of freedom:

$$p_{\text{diagnostic}} = \Pr\left(T_{13} \geq \frac{2.9 - |e|}{\text{SE}(e)}\right).$$

They give 0.000024 static and 0.001438 mobile. These probabilities condition on the point predictions; they do not integrate predictor uncertainty or correct for model selection, and are not independent confirmatory evidence. The original normal-tail probabilities are superseded.

S5 Historical selection and stored predictor replicas

The historical audit reads 147 archived arms across selected OBDI02 and OBTC02 conditions. For OBDI02 mobile arms it retains a final source and evaluates the three nested inclusion levels in the main text. The widest level still requires at least one finite radius. The original key *no outcome-dependent threshold at all* is therefore overbroad. V2 labels it *at least one finite radius* and discloses final-source conditioning. The reference radius reproduces the frozen operator implementation; this is not a new validation of its full transition-kernel exactness.

SEAL01’s `_repair_replicates.npz` stores four arrays of 16 predictor means: mobile empirical, mobile Poisson, static empirical and mobile Poisson at doubled intensity. The archive stores radii; V2 converts them to residual percentages against the frozen reference radius before calculating means and standard deviations. No new replicas are sampled. The empirical-minus-Poisson difference is 0.414 ± 0.196 percentage points. It does not sustain the earlier large, opposite-sign birth-flux attribution.

S6 Programme boundaries and negative findings

The claim matrix binds contextual dispositions as well as positive results. MYQBD01 remains `EXISTING_Q_DATA_INSUFFICIENT__PROSPECTIVE_Q_CALIBRATION_REQUIRED`. Its scalar exposure result for a first birth is not a fully identified two-source spatial operator. Organiser-only ledgers do not supply descendant exposures.

PQEC01’s 128 worlds are developmental despite their raw provenance. FLCR01 withdraws prospective-confirmatory claims and does not reconstruct a held-out split after outcomes. Its founder rejection does not establish reproduction. Neither its candidate screen nor a two-centre classification is a qualified source-response operator.

The separate EXP-SC-IOM-00 report and erratum are pinned at `4282fc6ead915639711f5096c7825d3880a640d4`. The erratum retains low-dimensional causal experience memory and **INDIVIDUATION: FAIL**; it reduces the storage claim to high-dimensional internal variation without established high-dimensional history information. Only these documentary boundaries are used. This architecture is not part of the 28-arm dataset. Route E, the persistence papers, PMCR01 and H13-SHIELD supply no positive claim or run.

S7 Reproduction and limits

From the paper directory, run `pythoncode/reproduce.py`. This verifies the pinned source manifest, recomputes raw summaries, emits numerical macros, tables and four figures, and checks claim bindings. With Tectonic available, `pythoncode/reproduce.py--tectonic/path/to/tectonic` also rebuilds both PDFs. Python requirements and exact audited versions are supplied.

The source manifest is an input; the audit never refreshes it to make failing hashes pass. The package includes existing raw data used by the figures. PQEC01’s larger raw package remains external. Its 128-file hash and schema audit is included and can be repeated using `audit_final.py--pqec-raw` with the extracted raw directory. Without that directory, a new build does not claim to have reverified those external bytes.

The final-lattice radius and digest are checked for every fresh arm. Intermediate radii are checked against stored summaries because full intermediate lattices are not archived here. No independent engine replication, parameter sweep, manuscript peer review, publication, DOI or merge is claimed.